INDEPENDENT CASE STUDY 2026

EUDR

Supply Chain & Financial Risk Assessment

INDEPENDENT SUSTAINABILITY BUSINESS ANALYSIS

Simulated due diligence model examining supply-chain traceability, evidence quality, sourcing exposure and financial sensitivity.

PREPARED BY

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Disclaimer

This is an independent case study developed for educational and professional portfolio purposes. All associated suppliers, production plots, shipment records and financial scenarios are fictional and based on simulated data.

The analysis draws on publicly available regulatory guidance and secondary research, with relevant sources acknowledged throughout. The findings and recommendations are illustrative and do not represent an assessment of any real organisation or supplier.

This project is not affiliated with or endorsed by any organisation referenced and does not constitute legal advice, an independent audit or certification of EUDR compliance.

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EXECUTIVE SUMMARY

An EU importer can have complete purchasing records and still lack a defensible account of where its palm oil was grown. This is the operational gap examined in this case study: the difference between knowing what a business bought and being able to demonstrate the origin and risk position of the material it intends to place on the EU market. EC guidance

Meridian Foods Europe B.V. is a fictional food manufacturer sourcing refined palm oil from Indonesia. The assessment links three simulated sourcing routes, twelve production plots and three planned shipments in an Excel decision model. Rather than whether a document marked “available” establishes readiness, it asks whether each record is usable, whether it can be traced through the collector, mill and refinery batch, and whether an unresolved issue changes the importer’s decision.

The distinction is material. Of sixty simulated evidence records, fifty-five are marked available (91.7%), yet only seven of twelve plots (58.3%) pass the initial documentary checks. Under segregated refinery processing, the remaining exceptions affect two planned shipments, or 46 of 70 tonnes (65.7%). If the refinery instead draws every output batch from a fully mixed input pool, exposure increases to all 70 tonnes. The second case is a stress test, not an assertion that blending occurred.

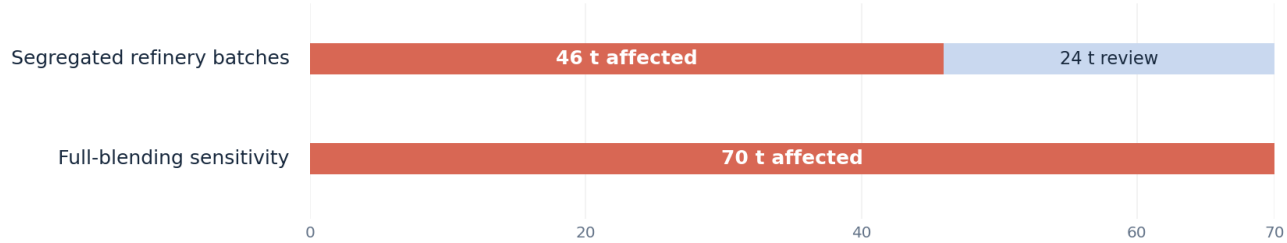


Figure 1. Planned volume affected by unresolved source evidence under two explicitly modelled refinery assumptions. Source: linked Excel model; all quantities are fictional.

The commercial sensitivity is also visible. For the 46-tonne segregated case, the combined low, middle and high assumptions produce €1,656, €5,474 and €12,420 respectively. These figures apply only where both assumed storage costs and an alternative-supply premium are incurred; they are neither observed P&L losses nor forecasts. Their value is to identify which commercial inputs management would need before choosing between delay, remediation and substitute supply.

The recommended decision is to retain an internal hold on the two affected shipments, pursue evidence-specific remediation and evaluate the outstanding substantive risks before any market-placement conclusion. An initial documentation pass does not satisfy Article 10: the operator must also assess relevant deforestation history after 31 December 2020, legality and land rights, source reliability and mixing risks, drawing on credible evidence. The workbook demonstrates the decision rules; it does not verify those underlying facts. EU law · EC guidance

THE BUSINESS CHALLENGE: FROM SOURCING TO TRACEABILITY

Refined palm oil is several transactions removed from its production plots. Fruit may move through collectors before milling, and crude oil from different mills may enter a shared refinery. At each transfer, an ordinary procurement record can remain commercially useful while losing some of the detail needed to attribute a finished product to land. That loss of connection, rather than paperwork volume in isolation, is the practical problem.

The European Forest Institute's 2026 preparedness exercise, IDH's Global Palm Oil Outlook and Trase's mapping of Indonesian export flows show why origin evidence, processing links and smallholder inclusion matter in practice. They inform the case design; none provides a finding about Meridian Foods or its fictional suppliers. EFI · IDH · Trase

The model follows four plots on each of Route's A, B and C through a collector and mill to refinery REF-01. Each route is associated with a separately identifiable output batch and a planned EU shipment. This permits an initial segregation case, while the alternative blending case tests the consequence if the refinery cannot substantiate separation.

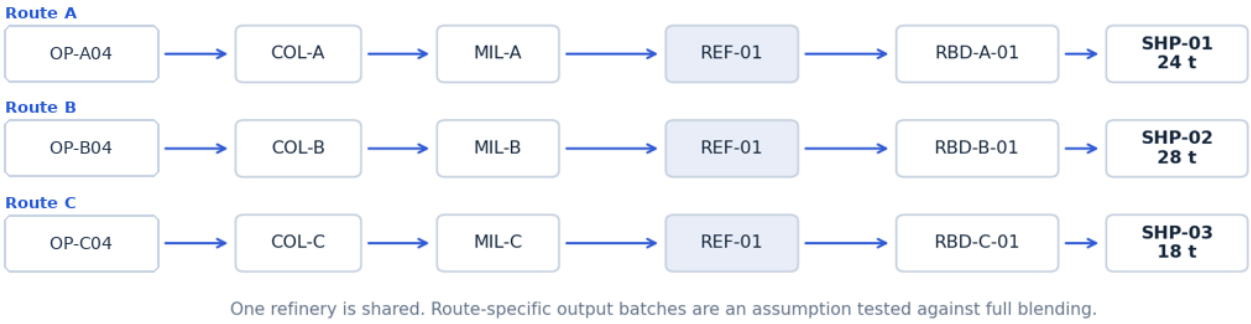


Figure 2. Physical supply route and modelled batch identifiers. The refinery is shared; the batch-segregation assumption is tested separately. Source: simulated supply-chain register.

Management ultimately needs to answer a narrower question than "is this supplier sustainable?": which shipment can move to further review, which must remain on hold, what missing information would change that position, and whether the resulting supply interruption warrants a contingency purchase. The report is organised around those decisions.

EUDR REQUIREMENTS AND BUSINESS IMPLICATIONS

The EUDR applies cumulative conditions to products in scope: relevant goods must be deforestation-free, legally produced in the country of production and covered by the required due diligence process. The fictional importer is assumed to be a medium/large upstream operator importing non-chemically modified RBD palm oil under Annex I heading 1511. On that basis, the 30 December 2026 application date applies. Indonesia is classified as standard risk, so the simplified low-risk route is not assumed. EU law · EU dates · Country list

The legal test begins with information but does not end there. Article 9 requires product and production-origin information, including geolocation for every contributing plot. Article 10 then requires an assessment of the risk that the product fails the Regulation: for example, whether the land was deforested after the cut-off, whether local legal and tenure requirements were met, whether the documents can be trusted and

whether material of unknown origin may have been mixed into the batch. Where risk is not negligible, Article 11 requires mitigation. These are related but separate decisions. EU law · EC guidance

Legal requirement	Business-analysis translation	Model boundary
Article 9: origin information	Associate every product batch with relevant plots, quantities, suppliers and evidence.	Tests linked IDs and structural completeness; cannot authenticate suppliers.
Article 10: risk assessment	Review forest history, legal rights, source credibility and mixing risk before making a substantive conclusion.	Records criterion status and outstanding evidence; does not certify negligible risk.
Article 11: mitigation	Keep material on hold while critical questions remain; define evidence needed to close each issue.	Tests whether a corrected record changes the internal decision.
Articles 8 and 12: due diligence system	Retain records, responsibilities and a reviewable decision trail.	Prototype registers only; no legal statement submitted.

The geolocation rule is particularly specific: plots greater than four hectares require polygons, while qualifying smaller plots may be represented by a point. Our model checks whether geometry type, coordinates and calculated area are internally consistent. It cannot determine whether a supplied polygon truly represents the farm. Nor does a production start date after the 2020 cut-off establish deforestation; it signals that historical land-use evidence deserves investigation. EU law · EC guidance

Indonesian production legality is more than a certification label. Presidential Regulation No. 16 of 2025 provides important context for the national sustainable palm-oil certification system, but the EUDR legal-production assessment still needs evidence relevant to the land, producer and applicable laws. Indonesia

Assessment Approach and Methodology

The method works backwards from a shipment decision. For each planned batch, the model identifies contributing plots, looks up collector and mill relationships and checks five evidence records per plot. An open exception is then carried into every linked shipment. This approach makes it possible to ask not only what is missing, but precisely which commercial decision the missing item prevents.

Five dimensions structure the initial review: availability, relevance, structural validity, consistency and traceability. They are deliberately not treated as five independently verified properties. Availability is taken from the simulated record; relevance and traceability are tested against identifiers; validity and consistency use geometry, coordinate and quantity rules. Authentication of a supplier document, verification of its legal meaning and verification of satellite land-use history remain outside the simulated model.

The additional data-quality screen checks blank or non-numeric coordinates, likely latitude/longitude reversal, an Indonesia bounding-box test, polygon direction and area tolerance, and overlapping model polygons. Batch quantities are reconciled to allocated plot tonnage. The coarse country screen is a plausibility check, not a confirmation of geographic location or legal parcel boundaries.

Risk criteria are recorded separately as Documented, Unresolved or Not screened, each with a reason. The Commission's Indonesia benchmark can be documented from a public source; plot-level deforestation, tenure and other substantive matters remain unresolved or unverified where no suitable evidence exists. This avoids the false precision of a composite compliance score. Country list · EC guidance

Finally, the workbook is challenged with input changes rather than judged only from its initial totals. A single missing geometry is corrected, two source exceptions are resolved together, a plot is moved to a different batch, an unresolved land-use indicator is changed, and the refinery setting is switched to blended. Those tests reveal whether the relationships and decision rules behave as intended; they do not establish that the fictional source information is true

Supply Chain Traceability and Evidence Review

The three planned shipments total 70 tonnes: SHP-01 (24 t), SHP-02 (28 t) and SHP-03 (18 t). The 12 plots produce 60 evidence entries, of which 55 are assigned an available status in the test dataset. The five remaining exceptions were chosen to represent different failure modes, then the linked model determines their downstream consequences.

Source	Specific exception	Shipment implication
B03 / EVD-031	Usable supplier geolocation is missing.	SHP-02 cannot establish complete source evidence.
B04 / EVD-037	Plot-to-mill custody record does not reconcile.	SHP-02 remains on hold even if B03 is corrected.
C02	Legality-related documentation is insufficient.	SHP-03 requires targeted document review.
C03	Production-period records conflict.	SHP-03 needs a reconciled source explanation.
C04	Land-use indicator remains unresolved.	SHP-03 requires enhanced substantive review.

The exceptions are concentrated in Routes B and C. All four Route A plots satisfy the initial documentary and structural tests, but that merely removes one barrier: the Article 10 risk assessment remains outstanding for SHP-01 too. Route B contains two information gaps. Route C contains two gaps and one unresolved indicator. These are internal assessment categories, not findings that real producers have deforested land.

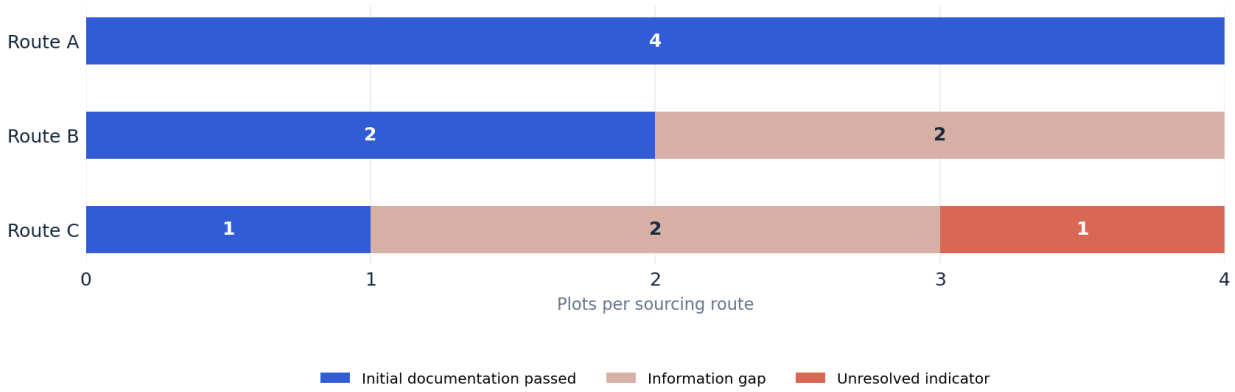


Figure 3. Plot-level outcomes produced by the simulated evidence and geometry checks. Counts describe initial documentation, not legal compliance.

The important result is the change of unit of analysis. An isolated record can look minor in a register; once it is connected to a contributing plot and refinery batch, it may hold an entire shipment. The link table makes this exposure visible, rather than assigning a batch status manually.

Key Findings, Risk Exposure and Financial Implications

The figures tell different parts of the same story. At record level, availability is 55/60 (91.7%). At plot level, only 7/12 (58.3%) meet the initial evidence criteria. Once those plot outcomes are linked to shipments under segregation, 46/70 tonnes (65.7%) are affected. The percentages should not be interpreted as a single attrition rate: each has a different denominator. Taken together, however, they show why a high volume of documents is a poor substitute for source-level decision readiness.

A worked decision: SHP-02

SHP-02 is attributed to batch RBD-B-01 and plots B01–B04. The B03 geolocation evidence is absent, and B04's custody record does not substantiate the complete plot-to-mill connection. Because both plots contribute to the same batch, the importer cannot rely on the other two plots' records to close the gap. The defensible internal response is therefore to hold the 28-tonne shipment while those issues are investigated.



Figure 4. SHP-02 decision trail. The hold is driven by source exceptions and recalculated through the plot-to-shipment mapping.

The workbook tests that reasoning. Supplying a valid simulated geometry and marking B03's evidence as available does not lift the hold: B04 remains unresolved. Closing both exceptions moves the shipment from documentary hold to "Pending Article 10", not to legal clearance. In another test, changing B03's batch key causes SHP-01 to inherit the open exception. This is the clearest proof that the model follows relationships rather than a pre-written conclusion.

The same discipline applies to SHP-03. Resolving C04's indicator reduces the count of land-use warnings, yet documentary issues at C02 and C03 still affect the shipment. That differentiation matters because a missing document and an unresolved deforestation-related indicator demand different kinds of investigation. Neither should be relabelled as a confirmed environmental violation without supporting evidence.

Seven executed tests, including a swapped-coordinate input and a blended commercial case, are recorded in the workbook. All seven produced the expected internal control response. The result validates the prototype's rule behaviour under those tests; it does not verify the factual accuracy of the inputs.

Evaluating Options and Stress-Testing the Supply Chain

The preferred responses differ by the cause of the exception. For SHP-02, the evidence gap is specific and potentially recoverable: request geolocation and custody reconciliation, retain the internal hold and reassess the full batch when both records are satisfactory. Immediate supplier replacement would carry an operational consequence without being compelled by the fictional evidence alone. For SHP-03, documentary remediation must be accompanied by enhanced land-use review and a contingency sourcing option if the concern cannot be resolved.

The shared refinery is the important counterfactual. If the business can demonstrate segregated input-to-output batches, only the B and C shipments inherit the open records: 46 tonnes. If the refinery instead creates a fully mixed input pool for all three outputs, the same five exceptions affect all 70 tonnes. The shift from 65.7% to 100% affected volume is a scenario result, not an observation about the refinery. It illustrates why reliable segregation records can materially change the scope of a sourcing restriction.

To make the consequence visible without inventing company accounts, the workbook applies three hypothetical cost scenarios to the affected tonnage. Low, mid and high cases combine storage duration, storage cost per tonne-day and alternative-supply price premium. At the baseline 46 tonnes, storage alone ranges from €966 to €9,660; the alternative premium alone ranges from €690 to €2,760. If both apply to the same volume, their combined sensitivity is €1,656, €5,474 and €12,420. Both components would not

necessarily arise in a real decision, so the combined range is explicitly conditional rather than a P&L forecast.

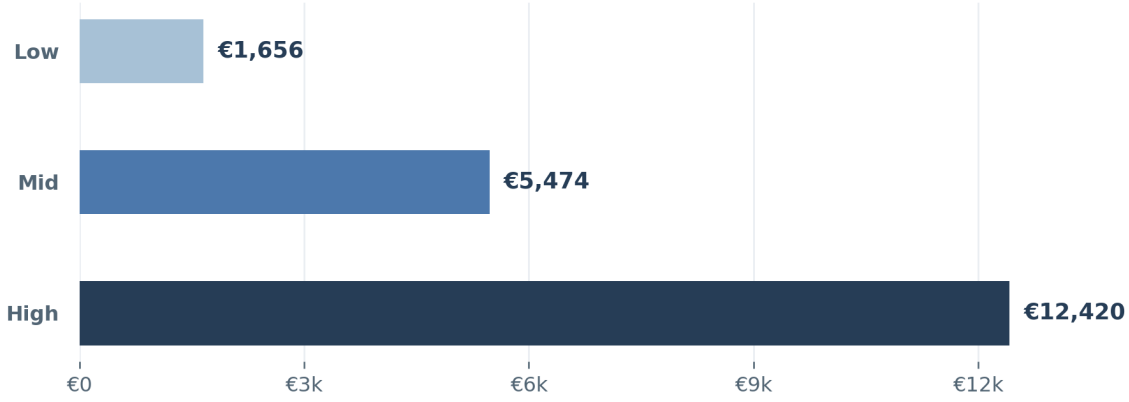


Figure 5. Illustrative combined cost sensitivity at 46 t: low €1,656, mid €5,474, high €12,420. Assumes storage and substitute-supply premium both apply; not actual losses or forecasts.

This comparison shows what management should ask for next: stock cover, critical production dates, supplier response timing, actual storage rates, current alternative quotations and whether delayed and replacement material would both be handled. Without those inputs, a precise financial-loss estimate would be less credible than the transparent scenario calculation. The cost model updates if the refinery setting changes, allowing the commercial discussion to follow the sourcing exposure rather than a fixed figure.

Supplier engagement remains the default where deficiencies are genuinely remediable. Research on Indonesian smallholders' documents uneven access to traceability and certification arrangements, making disproportionate exclusion a relevant concern. That consideration does not reduce the operator's legal obligations; it helps explain why a controlled remediation route may be preferable to automatic supplier termination. Chandra · Ekaputri

Recommendations, Controls and Implementation Plan

The next step for Meridian Foods would be to turn the five exceptions into supplier-specific requests, each with an evidence standard that a reviewer can evaluate. A supplier response should close an issue only when the original record is corrected, the information is relevant to the correct plot, and the linked shipment has been reassessed.

Exception	Recommended action	Evidence required for closure
B03 / origin	Procurement obtains usable plot geometry.	Correct plot, valid format and credible source provenance.
B04 / custody	Collector and mill reconcile intake and delivery records.	Documented plot-to-mill-to-batch linkage.
C02 / legality	Compliance requests supporting production-legality evidence.	Review against applicable Indonesian legal requirements.
C03 / period	Supplier reconciles production-period discrepancy.	Consistent dates supported by source records.
C04 / land use	Sustainability team investigates the warning.	Reliable land-use evidence and a documented risk reassessment.

The core business rules are consequently straightforward: retain persistent plot, collector, mill, batch and shipment keys; keep source status separate from data-quality checks; reconcile quantity allocations; never clear a documentary gate while a contributing plot has a critical exception; and require an independent

Article 10 conclusion before any regulatory placement decision. The workbook executes the relational and structural rules and records the outstanding substantive ones. It does not submit a due diligence statement.

Procurement analysts and sustainability/compliance reviewers are the intended users. They need to trace an exception from its origin; see which shipments it affects and document the basis for changing a decision. Sourcing decision-makers need the consolidated view: volume affected, unresolved risks, possible alternatives and the remaining evidence needed before an action is defensible.

At production scale, the same logic would require controlled data intake and a database rather than a manually maintained spreadsheet. The workbook includes a brief architecture note distinguishing this tested prototype from possible future ingestion through Power Query or an ETL process. No automated supplier integration or legal-compliance product is claimed.

Conclusion and Limitations

The central finding is not that Meridian Foods has uncovered deforestation or illegal production. It is that apparently good evidence coverage can conceal a concentrated operational dependency. A missing origin record and an unreconciled custody link are enough to keep SHP-02 on internal hold; a different set of questions requires enhanced review for SHP-03. When refinery mixing is assumed, the impact widens to the entire planned volume. These are decisions generated by transparent rules, not assertions about real suppliers.

The case demonstrates a practical business-analysis contribution: translating regulation into information requirements, connecting source records to commercial products, challenging assumptions and explaining the decision that follows. Its strongest discipline is the distinction between what a spreadsheet can test and what still needs substantive evidence.

All company identities, plots, quantities, dates and evidence statuses are constructed for this case study. The model does not authenticate supplier claims, independently evaluate forests or land rights, or establish EUDR compliance. Those tasks remain necessary in a real due diligence process, but the prototype clarifies where the questions would arise, what evidence would be needed and how an importer should avoid acting before the answers are adequate.

APPENDIX / MODEL CHECKS AND SOURCE NOTES

The figures in the report correspond to the workbook's calculated baseline: 55/60 evidence entries (Dashboard B4–B6), 7/12 documented plots (Dashboard B7–B8), 46/70 tonnes affected (Dashboard B9–B11), and the three segregated commercial scenarios (Commercial G2:G4). Blended-mode sensitivity is controlled by Dashboard B3 and the Plot x Shipment membership matrix.

Model test	Observed outcome
B03 geometry source corrected	SHP-02 remains held while B04 is open.
B03 and B04 corrected	SHP-02 advances to Pending Article 10, not legal clearance.
Refinery mode changed to Blended	All 70 tonnes are affected by open exceptions.
B03 moved to batch A	SHP-01 inherits the exception.
C04 indicator closed	SHP-03 remains held for C02 and C03.
Coordinate pair reversed	Data Quality flags reversal; affected shipment changes.
Blended commercial scenario	Low combined sensitivity recalculates to €2,520.

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